Alexander Agung Raya

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WORK EXPERIENCE

Fullstack Web Developer Jan 2026 - June 2026

PT. Surabaya Autocomp Indonesia - Internship *Mojokerto, Indonesia*

- Built scalable backend services with Django for internal web applications.
- Managed PostgreSQL queries and schema updates to improve data reliability.
- Implemented REST API endpoints and authentication flows for core features.

Stack: Django,PostgreSQL, REST API, JavaScript

Game Developer Sep 2025 — Present

Ariverse Studio - Contract (Remote) *Indonesia*

- Developed gameplay systems and interactive mechanics using Unity and C#.
- Collaborated with designers and artists to deliver immersive game experiences.
- Optimized performance for desktop and mobile targets.

Stack: Unity, C#, Game Development

PROJECTS

G2W — Game to Website (g2w.alexanderar.com) 2026

- Built and shipped an open-source, TypeScript-first framework for browser games as a pnpm monorepo of 8 published npm packages: renderer-agnostic core, Effect-based runtime, PixiJS renderer, DOM/gamepad input, web audio, devtools, and a `create-g2w` CLI scaffold.
- Enforced architecture by tests — the core has zero runtime dependencies and the same conformance suite runs against WebGL, Canvas, and WebGPU backends in real headless browsers; 453/453 tests green in CI on every commit.
- Shipped a live Astro documentation site with Ctrl+K search, playground, and a headless-Chrome benchmark harness (60 fps up to 5,000 sprites) comparing against raw PixiJS, Phaser 4, and Three.js, deployed to Cloudflare Pages via CI.

Stack: TypeScript, pnpm Workspaces, PixiJS, Effect, Vitest, Astro, Cloudflare Pages, npm Trusted Publishing

LearnMap (github.com/AlexanderDev2004/Learmap-kitbuild) 2026

- Reimplemented the KitBuild Concept Map platform (Hiroshima University) as my bachelor thesis: goal-map editor, automatic kit generation, and automatic diagnosis of student concept maps for formative assessment.
- Implemented the thesis experiment as a first-class feature — segmenting goal maps into ordered kit segments with Paragraph and Depth-Topic models — plus quiz builder, assignments, and a research analytics dashboard with CSV export.
- Deployed end-to-end on Cloudflare Workers with a typed Effect service layer, Drizzle ORM on LibSQL/Turso, and a drag-and-drop concept map canvas.

Stack: TanStack Start, React, TypeScript, Effect, Drizzle ORM, LibSQL/Turso, Cloudflare Workers

LogBook Internship (logbook.alexanderar.com) 2026

- Built a deployed internship log book platform that replaces the manual Word/Typst workflow: weekly activities, AI text refinement with bring-your-own API keys, and PDF export in the exact official campus template.
- Designed a three-party signing flow (student, field supervisor, academic advisor) through private single-use links with drawn signatures and lock-on-sign rules.
- Compiled the final PDF in-browser with the Typst WASM compiler and metric-compatible fonts, deployed via Alchemy to Cloudflare Workers with Drizzle on Cloudflare D1.

Stack: TanStack Start, React, TypeScript, Effect, Drizzle ORM, Cloudflare D1, Typst WASM, Alchemy

Portofolio Alex (alexanderar.com) 2026 — Present

- Rebuilt my personal portfolio from scratch using TanStack Start, React 19, Tailwind CSS 4, and Bun, deployed to Cloudflare Workers via GitHub Actions CI/CD.
- Built a typed Markdown content pipeline for projects, blogs, and security reports, with live GitHub data (contributions and merged PRs) snapshotted at build time and refreshed on the client via TanStack Query.
- Automated build-time OG image generation (Python/Pillow) for every page, with per-route SEO metadata and a fully type-safe file-based router.

Stack: TanStack Start, React, Tailwind CSS, TypeScript, Bun, Cloudflare Workers

Decimatex (github.com/AlexanderDev2004/decimatex) 2026

- Built a Decision Support System (DSS) platform that automates decision analysis using MCDM (Multi-Criteria Decision Making) methods.
- Developed a user-friendly interface for calculating and making decisions with dashboard monitoring.
- Created cross-platform desktop application using Electron with Bun, Hono, and PostgreSQL.

Stack: Bun, Tanstack, Effect, Hono, Drizzle, PostgreSQL, Tailwind CSS, Electron

GitTrace (github.com/ekyaaa/git_trace) 2026

- Developed a Flutter Desktop application that generates monthly internship reports from local Git activity.
- Built interactive calendar visualization with commit statistics and working hours tracking.
- Implemented multi-format export (Excel, Word, PDF) with custom template support and CI/CD automation.

Stack: Flutter, Dart, Git

Lapor Sana (github.com/AlexanderDev2004/LaporSana) 2025

- Built a campus facility issue reporting and management platform.
- Developed recommendation workflow support with Python-based decision logic.
- Implemented backend and UI modules for report handling and status tracking.

Stack: Laravel, MySQL, Tailwind CSS, Python

Ijen Driver (github.com/AlexanderDev2004/ijen-driver) 2025

- Developed a booking platform for tour driver services.
- Built full-stack features for booking data, driver data, and responsive pages.
- Improved usability with a simple flow and mobile-friendly interface.

Stack: Laravel, MySQL, Tailwind CSS

EDUCATION

State Polytechnic of Malang Malang

Diploma 4 Informatics Engineering *2023 — Present*

- Relevant focus: software engineering, web development, and databases.